

Algebra II with Trigonometry

Chapter 5.2

Olympiad Solutions

Saved tutoring response with MathJax rendering. Original PDF file:
[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Solutions

1. Without using a calculator, evaluate $\sin 75^\circ + \sin 15^\circ$ and explain any surprising symmetry in your answer.

Solution:

Recall the sum-to-product identity:

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

Set $A = 75^\circ, B = 15^\circ$:

$$\sin 75^\circ + \sin 15^\circ = 2 \sin \left(\frac{75^\circ + 15^\circ}{2} \right) \cos \left(\frac{75^\circ - 15^\circ}{2} \right) = 2 \sin(45^\circ) \cos(30^\circ)$$

We know:

$$\sin 45^\circ = \frac{\sqrt{2}}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}$$

So,

$$2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{2}\sqrt{3}}{2} = \frac{\sqrt{6}}{2}$$

Surprising symmetry: Note that $75^\circ + 15^\circ = 90^\circ$; i.e., the two angles sum to a right angle, yet their sines sum to the (positive) product of "means" sines and cosines. This comes from the general identity:

$$\sin x + \sin(90^\circ - x) = 2 \sin 45^\circ \cos(x - 45^\circ)$$

2. Given that $\cos \theta = \frac{12}{13}$ for an acute angle θ , determine $\tan(2\theta)$ mentally, using identities and minimal computation.

Solution:

First, $\sin^2 \theta = 1 - \cos^2 \theta = 1 - \frac{144}{169} = \frac{25}{169}$, so $\sin \theta = \frac{5}{13}$ (since θ is acute, sine is positive).

Recall:

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

But

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{5/13}{12/13} = \frac{5}{12}$$

So

$$\tan(2\theta) = \frac{2 \cdot \frac{5}{12}}{1 - \left(\frac{5}{12}\right)^2} = \frac{\frac{10}{12}}{1 - \frac{25}{144}} = \frac{5}{6} \div \frac{119}{144} = \frac{5}{6} \cdot \frac{144}{119} = \frac{120}{119}$$

3. Compute $(\cos 60^\circ + \cos 120^\circ + \cos 240^\circ)^2$ without a calculator.

Solution:

Compute each cosine:

$$\cos 60^\circ = \frac{1}{2}$$

$$\cos 120^\circ = \cos(180^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$$

$$\cos 240^\circ = \cos(180^\circ + 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$$

Sum:

$$\frac{1}{2} + \left(-\frac{1}{2}\right) + \left(-\frac{1}{2}\right) = -\frac{1}{2}$$

Now square:

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$$

4. The sum $S = \tan 15^\circ + \tan 45^\circ + \tan 75^\circ$ is an integer. Find its value, showing any clever simplification.

Solution:

Let's use the identity (for angles x, y, z with $x + y + z = 180^\circ$):

$$\tan x + \tan y + \tan z = \tan x \tan y \tan z$$

But $15^\circ + 45^\circ + 75^\circ = 135^\circ$, **not** 180° . Let's try direct calculation:

Compute each:

$$\tan 45^\circ = 1$$

$$\tan 15^\circ = 2 - \sqrt{3}, \quad \tan 75^\circ = 2 + \sqrt{3}$$

So,

$$S = (2 - \sqrt{3}) + 1 + (2 + \sqrt{3}) = 2 - \sqrt{3} + 1 + 2 + \sqrt{3} = 2 + 2 + 1 + (-\sqrt{3} + \sqrt{3})$$

Thus, $S = 5$.

5. Let $x = \sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ$. Find the exact value of x , using symmetry or patterns among the special angles.

Solution:

Compute each:

$$\sin^2 30^\circ = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\sin^2 45^\circ = \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{2}{4} = \frac{1}{2}$$

$$\sin^2 60^\circ = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$$

Sum:

$$x = \frac{1}{4} + \frac{1}{2} + \frac{3}{4} = \frac{1+2+3}{4} = \frac{6}{4} = \frac{3}{2}$$

6. For positive real numbers a and b , the triangle with vertices $(0, 0)$, $(a, 0)$, and (a, b) has angle θ at the origin. If $\sin \theta = \frac{b}{\sqrt{a^2+b^2}}$ and $\cos \theta = \frac{a}{\sqrt{a^2+b^2}}$, what is $\tan(2\theta)$ as a function of a and b ?

Solution:

We are given $\sin \theta = \frac{b}{\sqrt{a^2+b^2}}$ and $\cos \theta = \frac{a}{\sqrt{a^2+b^2}}$.

Recall:

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

But $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{b}{a}$.

Therefore,

$$\tan(2\theta) = \frac{2\frac{b}{a}}{1 - \left(\frac{b}{a}\right)^2} = \frac{2b/a}{1 - b^2/a^2} = \frac{2b}{a} \cdot \frac{a^2}{a^2 - b^2} = \frac{2ab}{a^2 - b^2}$$

Answer: $\boxed{\frac{2ab}{a^2 - b^2}}$